

Programme	B. Sc. Mathematics Honours			
Course Code	MAT6EJ306			
Course Title	INTRODUCTION TO FOURIER ANALYSIS			
Type of Course	Elective			
Semester	VI			
Academic Level	300-399			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	An introductory course in Real Analysis including series of functions			
Course Summary	Fourier analysis is a fundamental component in the tool-kit of every pure and applied mathematician with numerous applications to signal processing, image processing, tomography and several other areas of engineering. In this course we shall look at the most basic theoretical foundations of this subject. Along the way we shall have to recapitulate some of the requisite results from functional analysis.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate proficiency in defining and applying concepts related to inner product spaces, including orthogonality and linear operators.	Ap/An	P	Internal Exam/Assignment /Seminar/ Viva / End Sem Exam
CO2	Describe orthogonality, including definitions and examples. Demonstrate the use of orthogonal projections, including the Gram-Schmidt orthogonalization process.	Ap	C	Internal Exam/Assignment / Seminar/ Viva / End Sem Exam
CO3	Compute Fourier series on various intervals including cosine and sine expansions, and understand the complex form of Fourier series.	Ap	P	Internal Exam/Assignment /Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book		A First Course in Wavelets with Fourier Analysis, 2e, Albert Boggess and Francis J Narcowich, Wiley.		
Module	Unit	Content	Hrs (48+ 12)	Marks
				Ext: 70
I	Inner Product Spaces		12	
		Quick review through the preface of the text book for the discussions Fourier Analysis and Wavelets		
	1	0.1 and 0.2 – Motivation, definition and examples of inner product.		
	2	0.3 – The spaces L^2 and ℓ^2 – 0.3.1 - Construction of inner products in L^2 and ℓ^2 .		
	3	0.3.2 – Convergence in L^2 versus uniform convergence.		
	4	0.4 – Schwarz Inequality		
	5	0.4 - Triangle Inequality		
	6	0.5 – Orthogonality 0.5.1 – Definitions and examples.		
	7	0.5.2 – Orthogonal Projections – up to and including example 0.23		
II	Inner Product Spaces – contd.		12	
	8	0.5.2 – Orthogonal Projections – rest of the section		
	9	0.5.3 – Gram – Schmidt Orthogonalization.		
	10	0.6 – Linear Operators and their Adjoints 0.6.1- Linear Operators		
	11	0.6.2 – Adjoints - (up to and including Example 0.31)		
	12	0.6.2 – Adjoints – rest of the section.		
III	Fourier Series		12	

	13	1.1 – Introduction (1.1.1 to 1.1.3)		
	14	1.2 – Computation of Fourier Series 1.2.1 – On the interval $[-\pi, +\pi]$ – with examples		
	15	1.2.2 – Other intervals – with examples		
	16	1.2.3 – Cosine and Sine expansions with examples		
	17	1.2.5 – The complex form of Fourier Series		
	Modules III and IV are presented only for motivations and examples for the theory. All the proofs of theorems in these modules are optional to study and exempted from external examination.			
IV	Fourier Transforms		12	
	18	2.1 – Informal development of the Fourier transform 2.1.1 – Fourier Inversion Theorem		
	19	2.2.2 – Fourier Transform of a convolution		
	20	2.2.3 – Adjoint of the Fourier Transform		
	21	2.2.4 – Plancherel Theorem		
	22	More problems from the above sections		
V (Open Ended)	Fourier Analysis		12	
	After having the above basics of Fourier Analysis, one can look at the discrete versions of Fourier Analysis and can enter into Wavelets theory (for instance refer sections 4.1 and 4.2 of text book). The Haar wavelet analysis with its decomposition and reconstruction theorems open the window to signal theory which is an active area of research for both pure and applied Mathematicians			

References

1. Ten lectures on Wavelets, Daubechies, Philadelphia, SIAM, 1992.
2. Fourier Analysis and its Applications, Gerald B Folland, Wadsworth and Brooks/Cole Advanced Books and Software, Pacific Grove, California.
3. Introduction to Fourier Analysis on Euclidean Spaces, Elias M Stein and Guido -Weiss, Princeton University Press.
4. How to make Wavelets, Robert S. Strichartz, The American Mathematical Monthly.

Note: 1) Optional topics are exempted for end semester examination.

2) 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	3	2	1	3	1	3	1	3	0	1
CO 2	3	3	2	1	3	1	3	1	3	0	1
CO 3	3	3	2	1	3	1	3	1	3	0	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓